

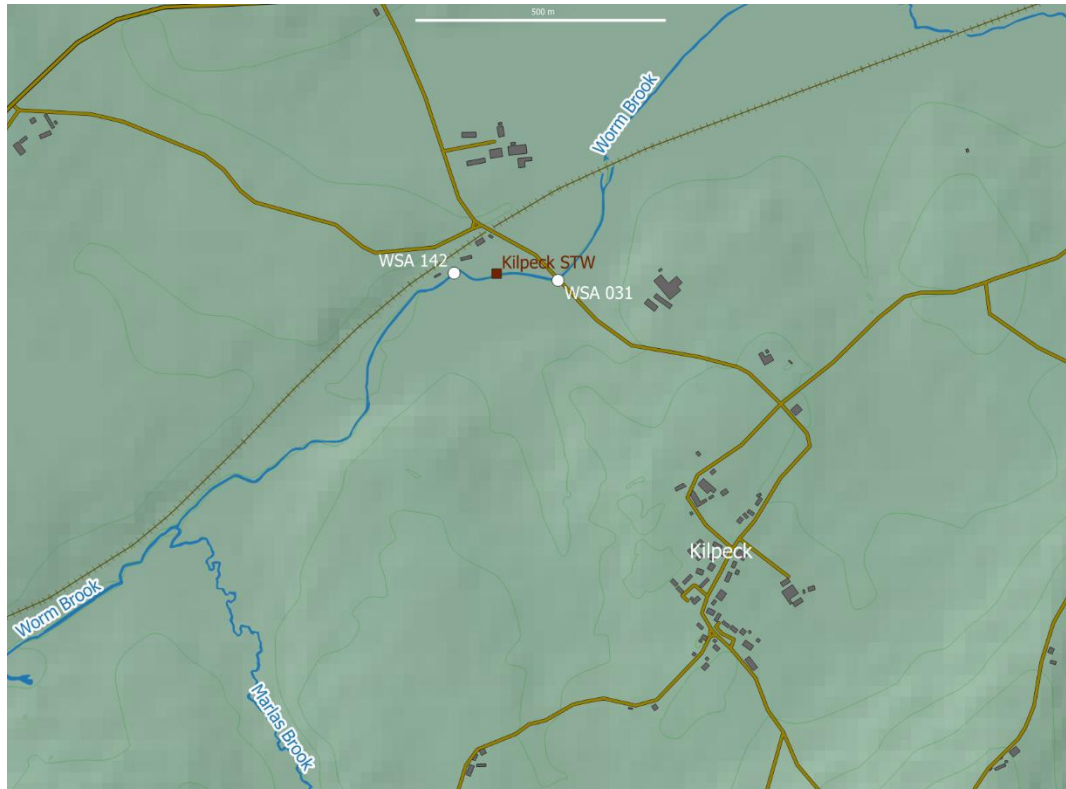


Hot-spot report: Worm Brook at Kilpeck

A report prepared on behalf of Wye Salmon Association (WSA)

Version	2.0	WSA hotspot ref.
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Location



Map of the area around Kilpeck and the adjacent Worm Brook. White circles are WSA sampling sites.

Kilpeck is situated close to the Worm Brook. The village is served by Kilpeck STW (operated by DC-WW; input load 85 PE), discharging into the Worm northwest of the village.

WSA sample the Worm at two sites, Goughs Bridge (WSA 031) & WSA 141, located upstream and downstream of the STW outflow.

STW malfunction

In addition to routine water quality measurements, during much of 2023, the outflow from Kilpeck STW has been observed discharging extremely dirty water. DC-WW were notified of the problem and took appropriate action. Jennifer Grubb of DCWW reported in an e-mail of 25-Sep-23 that: *"Kilpeck was completely emptied on the day that the issue was reported. We have since had two tankers attend the site every day to ensure there is no discharge from the site. On 21/09/2023 the operational teams completely emptied the site again and installed a new diffuser air blower, as the old one was blocked which had stopped aeration at the site, this is what was causing the issues with the discharge. The dissolved oxygen levels and new samples are all compliant with our permit and so the works will began*

discharging to the river from 22/09/2023. We will also have an operator visit every day to make sure we are in compliance range."

Beyond this malfunction, it is worth examining if Kilpeck STW has any impact on the Worm Brook more generally.

Hypothesis

That Kilpeck STW exports unacceptable levels of phosphorus and other pollutants into the River Dore.

Test

RP and EC signals from the two sites, WSA 031 & WSA 141, located upstream and downstream of the STW will be compared. Paired samples from these sites have been measured within 30 minutes of each other, approximately weekly, over the period Jan-23 to Aug-23.

Results

The results from 35 paired samples are summarised in the table below:

Parameter	Upstream (WSA 031)	Downstream WSA 141)	t-test
Mean RP (mgP/l)	0.18	0.16	49.18%
Mean EC @25°C (µS/cm)	659	659	88.24%

Mean values for reactive phosphorus and electrical conductivity on the Worm Brook at locations upstream and downstream of Kilpeck STW, during 2023. The 2-sided, paired t-test gives the probability that the two mean values belong to the same population (i.e., no change).

This tells us that the small drop in mean RP is probably not significant. Mean EC remains the same, with high confidence.

Conclusion

- Kilpeck STW has no statistically significant influence on RP levels in the Worm Brook.
- Similarly, there is no change in electrical conductivity.
- We have separately estimated that the discharge from Kilpeck STW represents approximately 0.4% of Q95 flow. This is quite small, and we would therefore not expect to see significant impact from Kilpeck STW in normal operation.
- It is nevertheless of concern that DCWW have no telemetry on their facility to notify them of a serious malfunction. This problem was only identified because WSA citizen scientists happen to monitor at this location. Surely DCWW can do better than this in 2023.